

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

192513077

I D. MOWLESHWARAN (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: mowleswaran

Level-3
10

Smart Home Automation System

understanding OF concepts.

A smart home controls lights, and temperature, and security using sensors and used preferences.

Explanation:

i) Simple Reflex Agent: Remembers rooms status and user habits.

ii) Turns light ON/OFF when motion is detected.

iii) Goal-Based Agent: keeps the home comfortable and secure.

iv) Utility - Based Agent: chooses the best action to save money and improve comfort

Application

A Hybrid Agent (model - Based + utility - Based) is the best choice because it learns user behavior and selects the most efficient action.

organization.

i Detect sensor data.

ii Analyze user preferences.

iii choose user best action.

iv) control lights, Ac and security.

Accuracy

The hybrid agent provides comfort, security and energy savings.

clarity

A Hybrid Agent is the most effective solution.

2) Robot in an unknown maze

understanding of concepts

The robot must find the next without knowing the maze.

Explanation.

i) uniform cost search (ucs):

- * Finds the lowest-cost path
- * uses more memory.

ii) Iterative Deepening search (IDS):

- * uses less memory.
- * Repeats the search at deeper levels.

Application

uses a Goal-Based Agent with UCS because the robot's goal is to find the safest and shortest path.

organization.

i) start exploring

search possible paths.

iii) Avoid blocked paths.

iv) Reach the exit

Accuracy.

UCS gives the optimal path, while DS is better when memory is limited.

Clarity.

~~A~~ ~~GO~~

A Goal-Based Agent + UCS is the best combination.

3. Smart Traffic management System.

Understanding of Concepts.

The system controls traffic signals using vehicle density, pedestrians, and weather.

Explanation

i) Simple Reflex Agent: changes signals based on ~~current~~ traffic

ii) Model-Based Agent: Remembers traffic history.

iii) Goal-Based Agent: Reduces traffic congestion.

iv) Utility-Based Agent: selects the best signal timing to reduce waiting time.

Application:

The hybrid system reduces traffic jams and travel time.

Organization:

- i) collect traffic data.
- ii) Analyze road conditions.
- iii) Adjust road timing.
- iv) Reduce congestion.

Accuracy:

A Hybrid Agent (model-based utility-based) is the best choice because it learns traffic patterns and improves signal timing.

Clarity:

A Hybrid Agent is the most effective solution for smart traffic management.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand